

Proof of Elapsed Time (PoET) and Practical Byzantine Fault Tolerance (PBFT). Businesses can select the algorithm that best suits their needs in terms of security, scalability, and performance.

Hyperledger Sawtooth has some additional architectural features for enhanced performance.

- **Asynchronous client/server pattern:** To optimise scalability and performance, Sawtooth employs an asynchronous communication pattern between clients and validators. This approach allows for concurrent processing of requests, improving overall throughput.
- **Parallel transaction execution:** Sawtooth leverages a technique known as batching, where related transactions are grouped together. This enables efficient parallel processing of multiple transactions simultaneously, further enhancing the platform's performance.

Sawtooth validator

The validator in Hyperledger Sawtooth plays a pivotal role in the architecture of the blockchain network, acting as the core component responsible for the primary operations and integrity of the distributed ledger. Let's look at its main functions and characteristics.

Transaction processing

- **Validation and execution:** Validators are responsible for validating transactions against the network's consensus rules and executing them accordingly. They ensure that transactions are properly signed, correctly formatted, and adhere to the specific rules of their transaction families.
- **Interaction with transaction processors (TPs):** Validators communicate with transaction processors, which contain the business logic for different types of transactions. This communication is crucial for

executing the transactions as per the defined smart contract logic.

Block management

- **Block validation:** Validators verify the integrity and validity of each block before it's added to the blockchain. This includes checking the block's hash, ensuring all transactions within the block are valid, and confirming that the block adheres to the chosen consensus model's rules.
- **Consensus participation:** Validators play a central role in the consensus process, participating in the algorithm (e.g., PoET, Raft) that the network uses to agree on the state of the blockchain. This process ensures that only valid blocks are added to the chain and helps maintain the network's overall integrity and agreement on the ledger's state.

State management

- **State storage:** The state of the blockchain, which includes all balances, contracts, and relevant data, is managed by the validator. It uses a Merkle Radix Tree for efficient storage and querying of the blockchain state, facilitating quick verification and updates.
- **State transition:** Upon successful validation and execution of transactions, validators apply the changes to the blockchain's state. This process involves updating the current state based on the outputs of transactions, ensuring the ledger accurately reflects all executed transactions.

Networking and communication

- **P2P networking:** Validators use a peer-to-peer (P2P) network to communicate with each other, sharing blocks, transactions, and consensus-related messages. This network is fundamental for the distribution and verification of the ledger's data across all participating nodes.

- **Client interaction:** Validators expose interfaces (e.g., REST API) for clients to submit transactions, query the blockchain state, and retrieve information about blocks and transactions. This allows users and applications to interact with the blockchain network seamlessly.

Security and integrity

- **Cryptography:** Validators use cryptographic techniques to ensure the security and integrity of transactions and blocks. This includes verifying digital signatures, generating cryptographic hashes for blocks, and securing communications between nodes.
- **Permissions and policies:** Validators enforce network policies and permissions, controlling access and actions within the blockchain network. This ensures that only authorised participants can perform certain actions, enhancing the network's security.

The validator node is illustrated in Figure 2.

Transactions and batches

In Hyperledger Sawtooth, transactions are the fundamental elements through which the state of the blockchain is modified. Transactions are grouped into batches to efficiently process and confirm multiple operations at once. Understanding both concepts is crucial for developing applications on Sawtooth. Here's an overview.

Transactions

A transaction in Sawtooth represents an atomic change to the state of the ledger. Each transaction is composed of several key components.

- **Header:** Contains metadata about the transaction, such as the family name (which indicates the transaction processor that should handle the transaction), family version, and the inputs and